

BCNF

BCNF requires that for every functional dependency $A \rightarrow B$, A must be a superkey, meaning no partial dependencies or transitive dependencies can exist.

1. Farmer Table

- **Primary Key:** farmer_id
- **FDs:**
 - farmer_id $\rightarrow$ first_name, last_name, email, zip_code, contact_number, current_address, city, state, lat, long
- **BCNF Check:**
 - farmer_id is a superkey. **(BCNF)**

2. Expense_Category Table

- **Primary Key:** expense_category_id
- **FDs:**
 - expense_category_id $\rightarrow$ category_name, description
- **BCNF Check:**
 - expense_category_id is a superkey. **(BCNF)**

3. Expense Table

- **Primary Key:** expense_id
- **FDs:**
 - expense_id $\rightarrow$ expense_category_id, amount, date
 - expense_category_id $\rightarrow$ amount, date (if multiple expenses have the same category)
- **BCNF Check:**
 - expense_id is a superkey. However, if the second FD holds, it violates BCNF as expense_category_id is not a superkey. **(Not in BCNF)**

4. Farmer_Expense Table

- **Composite Primary Key:** (farmer_id, expense_id)
- **FDs:**
 - (farmer_id, expense_id) $\rightarrow$ (farmer_id, expense_id)
- **BCNF Check:**
 - This table is in BCNF as the primary key is a superkey. **(BCNF)**

5. Product Table

- **Primary Key:** product_id
- **FDs:**
 - product_id $\rightarrow$ product_name

- **BCNF Check:**
 - product_id is a superkey. **(BCNF)**

6. Product_Cost Table

- **Composite Primary Key:** (product_id, created_at)
- **FDs:**
 - (product_id, created_at) → price_per_bundle
- **BCNF Check:**
 - This is in BCNF. **(BCNF)**

7. Base_Market_Incentive Table

- **Composite Primary Key:** (product_id, date)
- **FDs:**
 - (product_id, date) → price
- **BCNF Check:**
 - This is in BCNF. **(BCNF)**

8. Harvest Table

- **Composite Primary Key:** (product_id, farmer_id, created_at)
- **FDs:**
 - (product_id, farmer_id, created_at) → quantity_in_bundles
- **BCNF Check:**
 - This is in BCNF. **(BCNF)**

9. Customer Table

- **Primary Key:** customer_id
- **FDs:**
 - customer_id → first_name, last_name, email, zip_code, contact_number, current_address, city, state, latitude, longitude
- **BCNF Check:**
 - customer_id is a superkey. **(BCNF)**

10. Orders Table

- **Primary Key:** order_id
- **FDs:**
 - order_id → customer_id, shipping_address, created_at
- **BCNF Check:**
 - order_id is a superkey. **(BCNF)**

11. Order_Detail Table

- **Primary Key:** order_detail_id
- **FDs:**
 - order_detail_id → product_id, order_id, quantity, total_price, created_at
- **BCNF Check:**
 - order_detail_id is a superkey. **(BCNF)**

12. Transactions Table

- **Primary Key:** transaction_id
- **FDs:**
 - transaction_id → order_id, order_status, timestamp
- **BCNF Check:**
 - transaction_id is a superkey. **(BCNF)**

13. Payment Table

- **Primary Key:** payment_id
- **FDs:**
 - payment_id → order_id, payment_method, amount_paid, payment_date
- **BCNF Check:**
 - payment_id is a superkey. **(BCNF)**